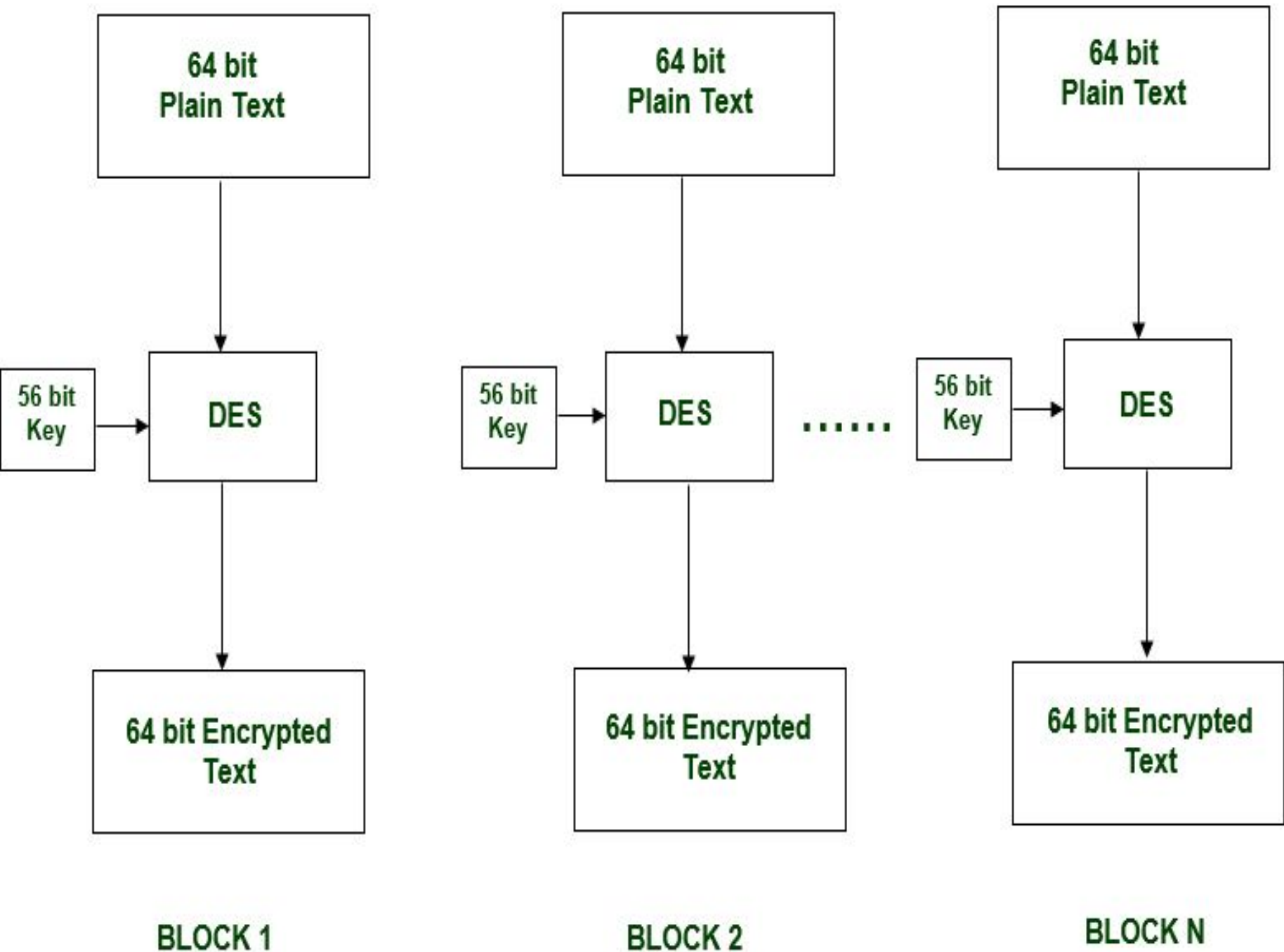


(Data encryption standard)

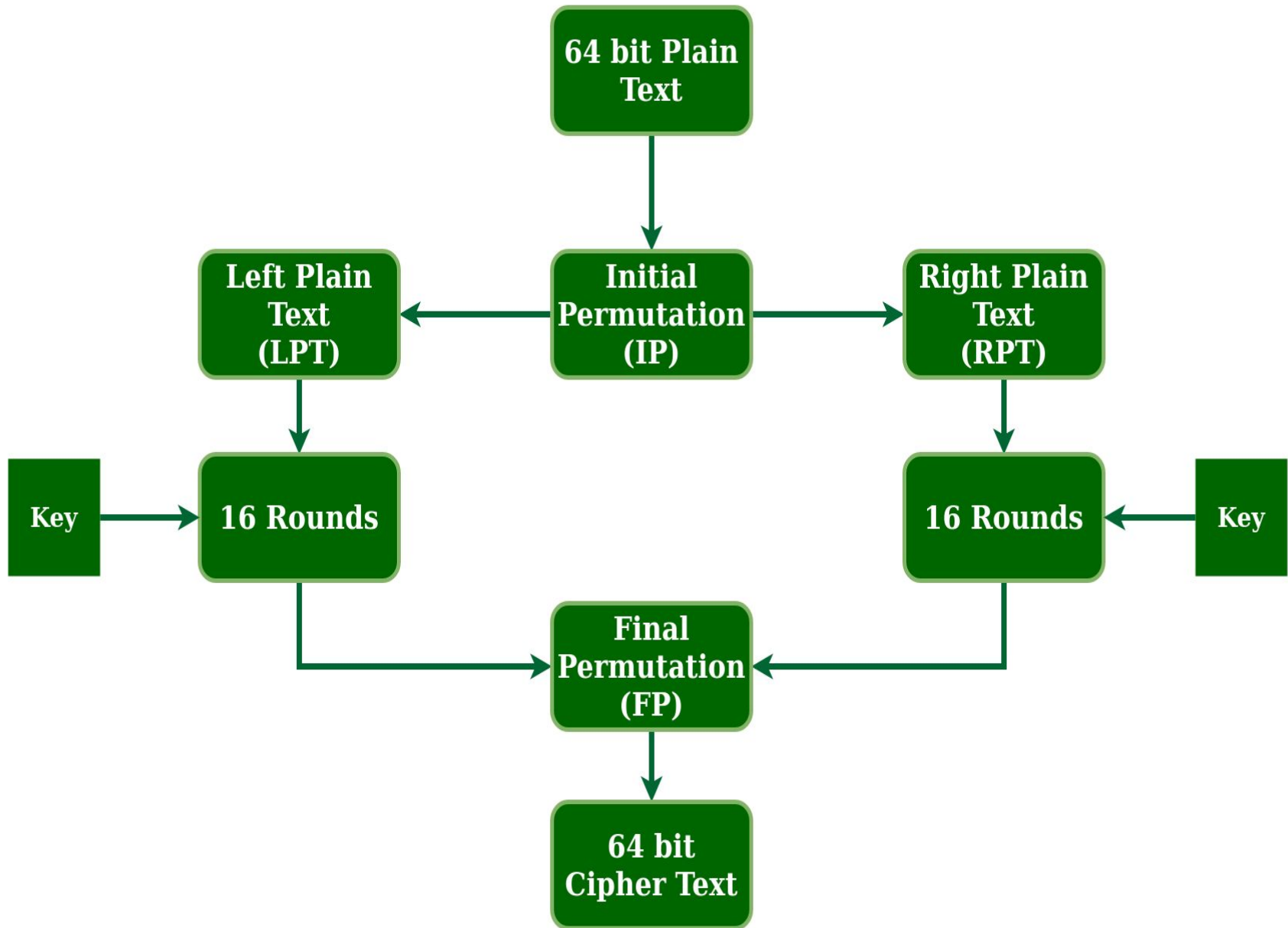
DES

- Data Encryption Standard (DES) is a block cipher with a 56-bit key length.
- It encrypts data in blocks of size of **64 bits** each, which means 64 bits of plain text go as the input to DES, which produces 64 bits of ciphertext. The same algorithm and key are used for encryption and [decryption](#),



Data encryption standard (DES)

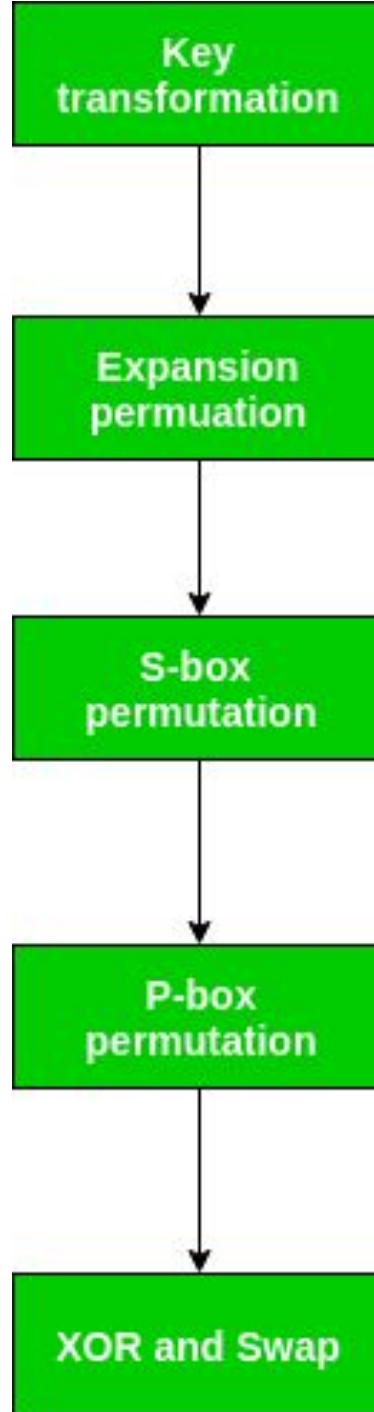
- DES is based on the two fundamental attributes of [cryptography](#): substitution (also called confusion) and transposition (also called diffusion).
- DES consists of 16 steps, each of which is called a round.
- Each round performs the steps of substitution and transposition.



Broad Level Steps in DES

Initial Permutation (IP)

As we have noted, the initial permutation (IP) happens only once and it happens before the first round.



Step 1: Key transformation

- initial 64-bit key is transformed into a 56-bit key by discarding every 8th bit of the initial key.
- Thus, for each a 56-bit key is available.
- From this 56-bit key, a different 48-bit Sub Key is generated during each round using a process called key transformation.

Step 2: Expansion Permutation

- after the initial permutation, we had two 32-bit plain text areas called Left Plain Text(LPT) and Right Plain Text(RPT).
- During the expansion permutation, the RPT is expanded from 32 bits to 48 bits.
- Bits are permuted hence called expansion permutation.
- This happens as the 32-bit RPT is divided into 8 blocks, with each block consisting of 4 bits.

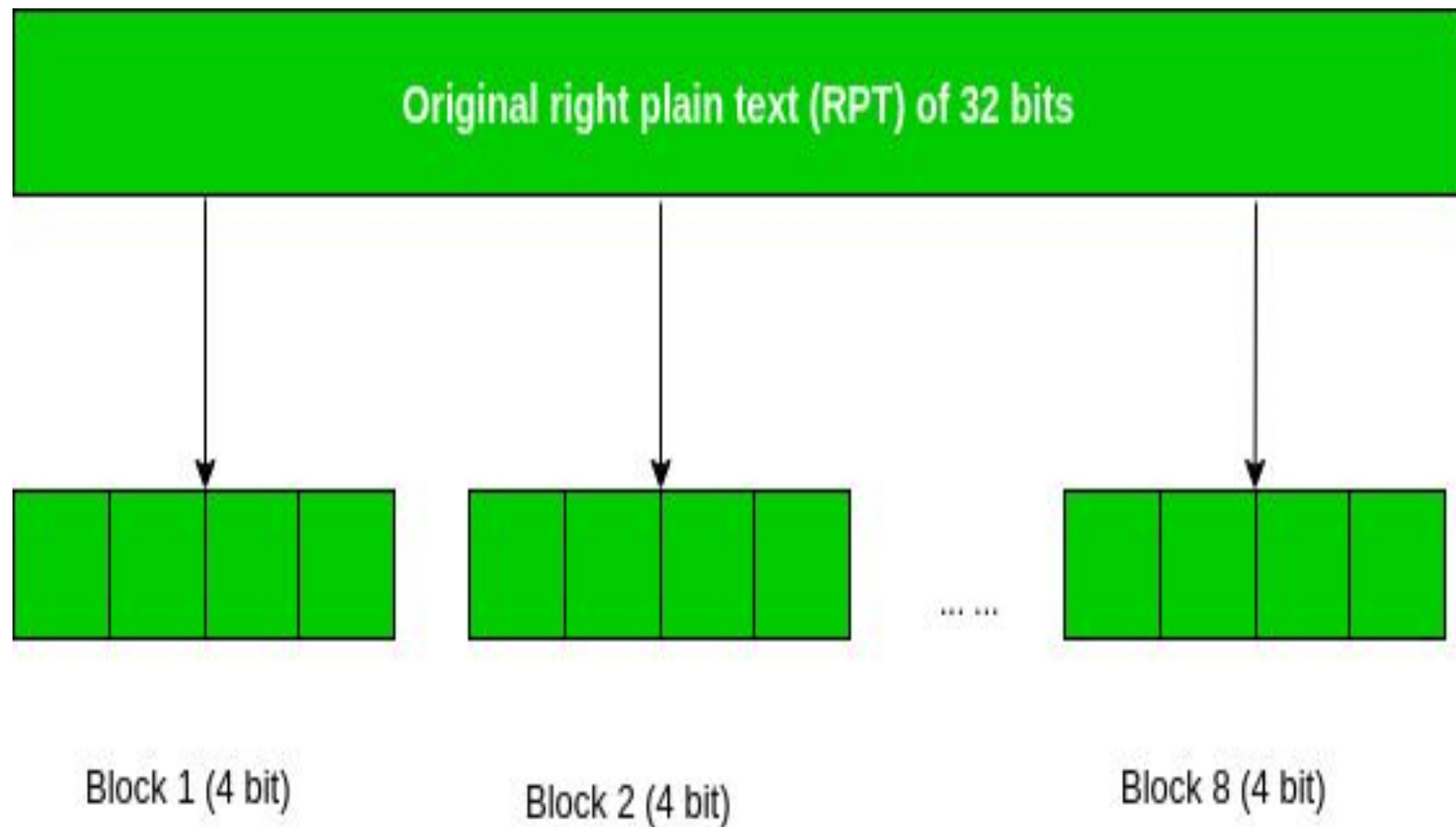
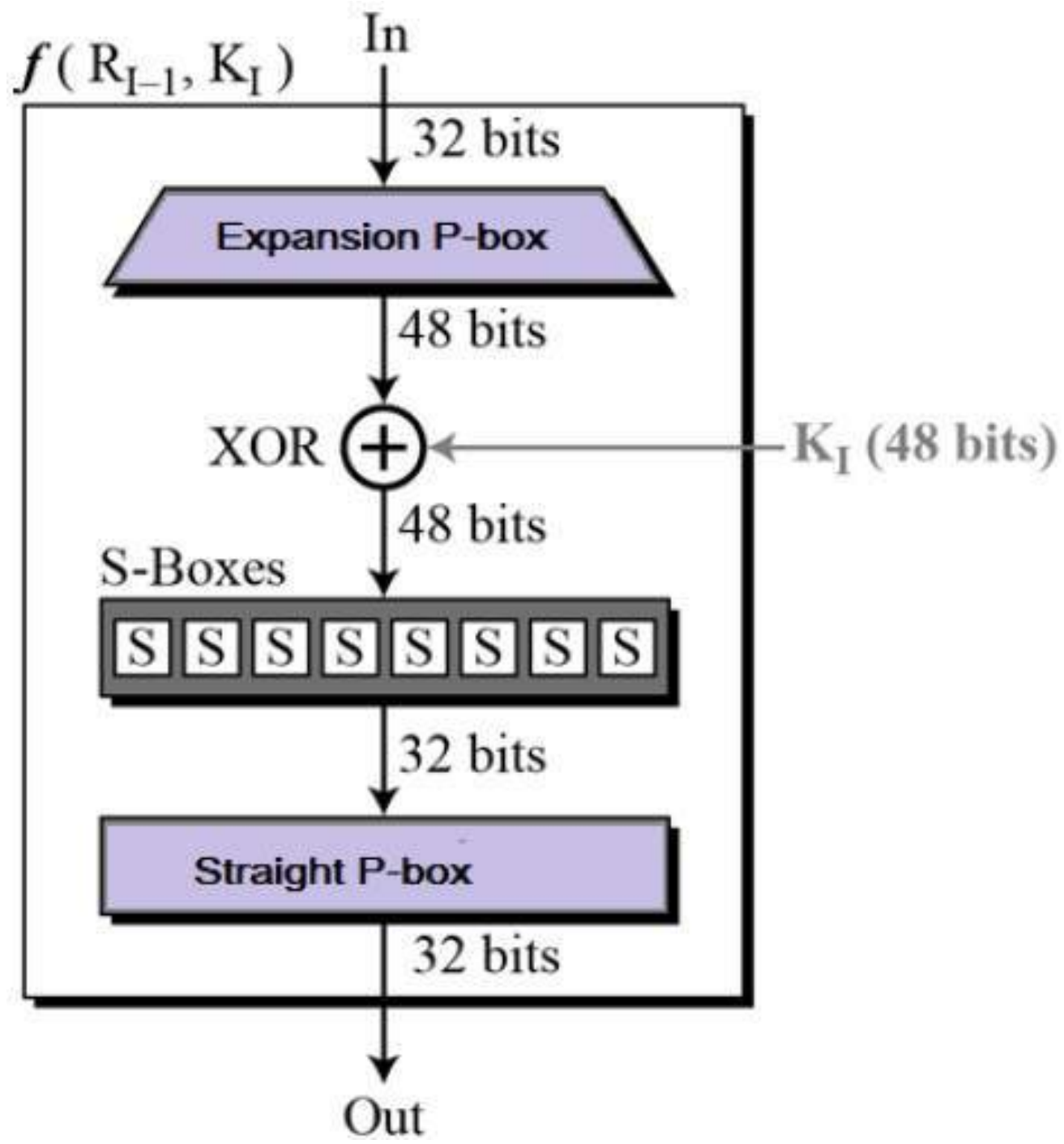


Figure - division of 32 bit RPT into 8 bit blocks

The key transformation process compresses the 56-bit key to 48 bits. Then the expansion permutation process expands the **32-bit RPT** to **48-bits**.

Now the 48-bit key is XOR with 48-bit RPT and the resulting output is given to the next step, which is the **S-Box substitution**.



S box Substitution

S box substitution is a process that accepts 48 bit input from the XOR operation and produces a 32 bit output using the substitution technique.

The substitution is performed by eight substitution boxes(also called as S boxes).

P box permutation

- The output of S-box consists of 32 bits.
- These 32 bits are permuted using a P box.
- Permutation mechanism involves replacement of each bit with another bit , without any expansion or compression.

XOR and Swap

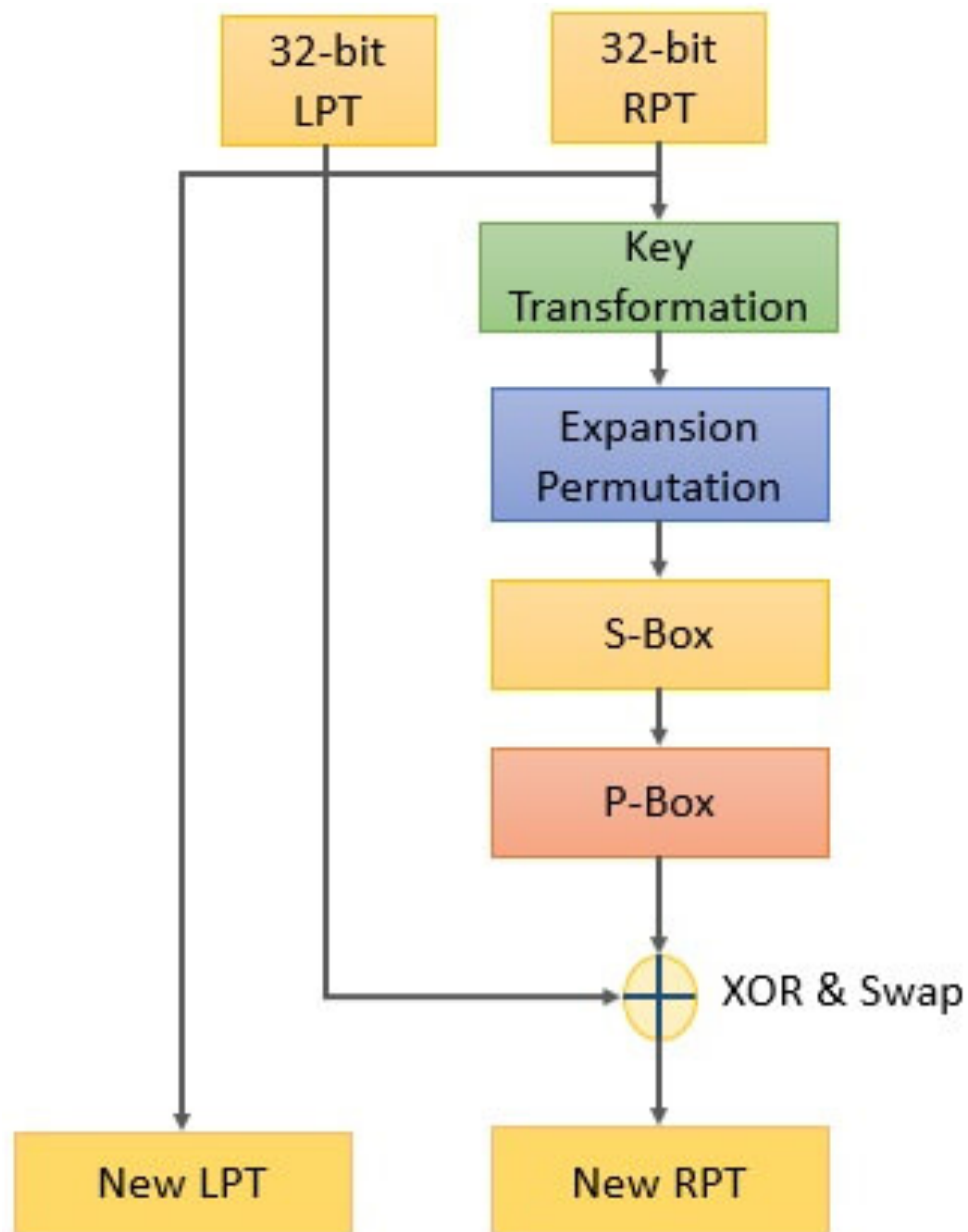
Note that we have been performing all these operations only on 32 bit right portion of 64 bit original plain text.

The left half portion was remain as it is.

In this step ,Left portion of initial 64 bit plain text is XORed with output produced by P.box permutation.

The result of this XOR operation becomes new right half(i.e. RPT).

The old right half becomes new left half in a process of wapping.



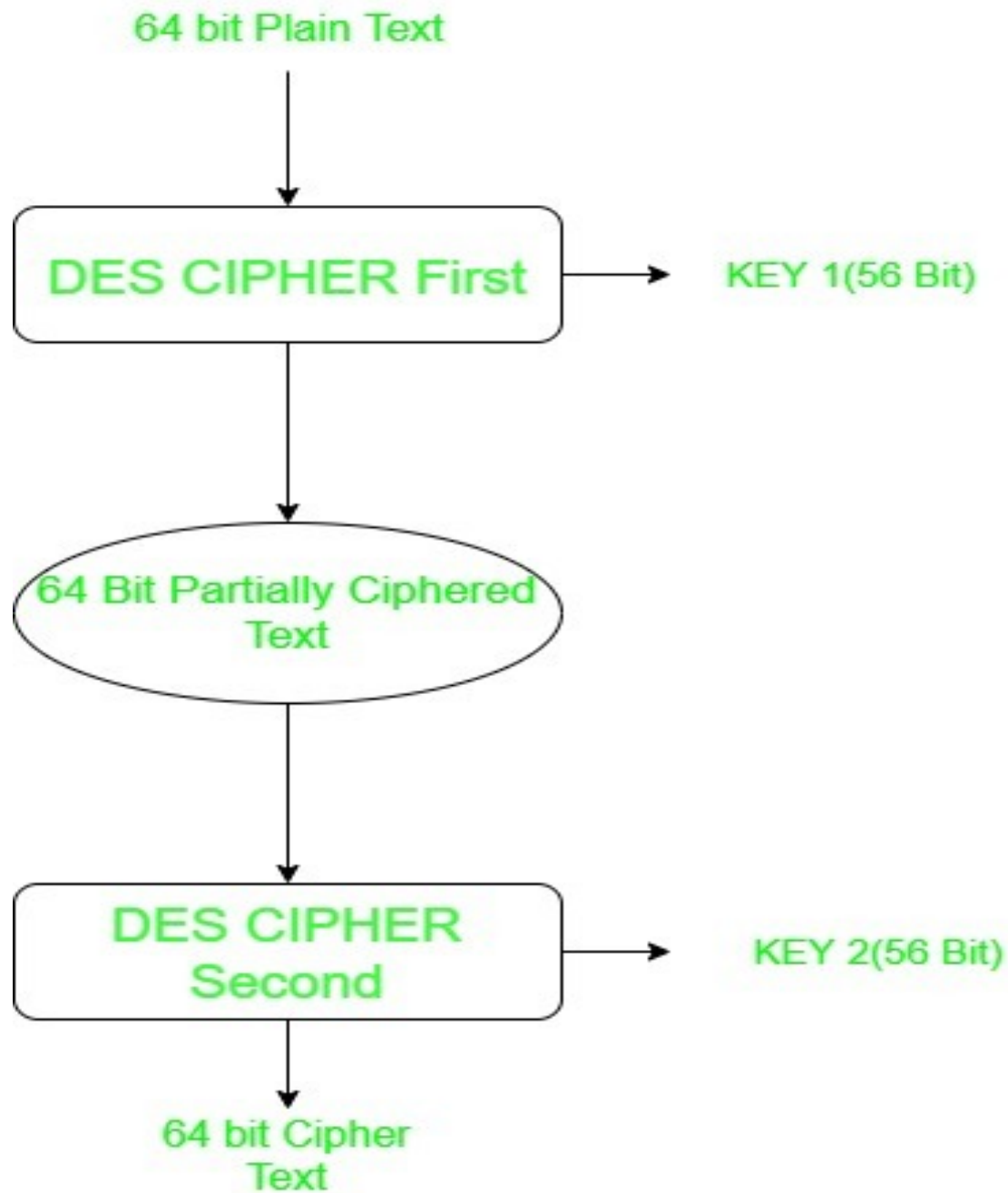
Final permutation

At the end of the 16 rounds, the Final Permutation is performed.

The output of final Permutation is the 64 bit encrypted block.

Double DES:

- Double DES is a encryption technique which uses two instance of DES on same plain text.
- In both instances it uses different keys to encrypt the plain text.
- Both keys are required at the time of decryption.
- The 64 bit plain text goes into first DES instance which then converted into a 64 bit middle text using the first key and then it goes to second DES instance which gives 64 bit cipher text by using second key.



Triple DES:

- Triple DES is a encryption technique which uses three instance of DES on same plain text.
- It uses there different types of key choosing technique in first all used keys are different and in second two keys are same and one is different and in third all keys are same.

